Havells Company Analysis

1. Project Overview

This project analyzes using transactional data from 20,000 purchases across various product categories. The goal is to analyze sales data for business insights, identify revenue trends over time, find top performing products and regions, understand customer payment behavior, and support data-driven business decision making

2. Project Overview

- Rows: 20,000
- Columns: 10
- Key Features:
- Customer demographics (Customer, Region)
- Purchase details (OrderID, Date, Product, Category, Quantity, Unit_Price, Revenue)
- Shopping behavior (Payment_Method)

3. Tools & Technologies Used

Python (Pandas, NumPy) – Data cleaning and analysis
MySQL – Data storage and querying
Power BI – Data visualization and dashboard creation
Excel – Data source format

4. Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was performed using Python to understand the dataset and extract key business insights. The Pandas library was used for data cleaning, analysis, and summarization.

● Data Loading: Imported the dataset using pandas.
● Data Inspection: Initially, the dataset was checked for structure, missing values, and data types.
● Revenue Analysis: Total revenue and distribution across products, Category and regions were analyzed.

```
Category
Accessories    922377326
Gadgets        919053750
Electronics    913920033
Name: Revenue, dtype: int64
```

```
Product
Tablet         407552365
Headphones     398626250
Phone          395918793
Printer        391311805
Watch          389094222
Laptop         387632824
Camera         385214850
Name: Revenue, dtype: int64
```

```
Region
West       567687927
North      550116667
Central    549787572
South      546973255
East       540785688
Name: Revenue, dtype: int64
```

● Payment Analysis: Payment behavior of customers was analyzed.

```
PaymentMethod
Net Banking    5077
Cash           5044
Card           5023
UPI            4856
Name: count, dtype: int64
```

.

● Database Integration: Connected Python script to SQL WorkBench and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

Total Revenue Analysis: This query calculates the total revenue generated from all transactions.

Total_Revenue
2755351109

Total Orders: Count the total Orders fulfilled by the company.

Total_Orders
20000

Revenue by Product: Identifies top-performing products based on revenue.

Product	Total_Revenue
Tablet	407552365
Headphones	398626250
Phone	395918793
Printer	391311805
Watch	389094222
Laptop	387632824
Camera	385214850

Revenue by Region: Shows which regions generate the highest sales

Region	Total_Revenue
West	567687927
North	550116667
Central	549787572
South	546973255
East	540785688

Payment Method Analysis: Shows customer payment preferences (UPI, Card, Cash, etc.).

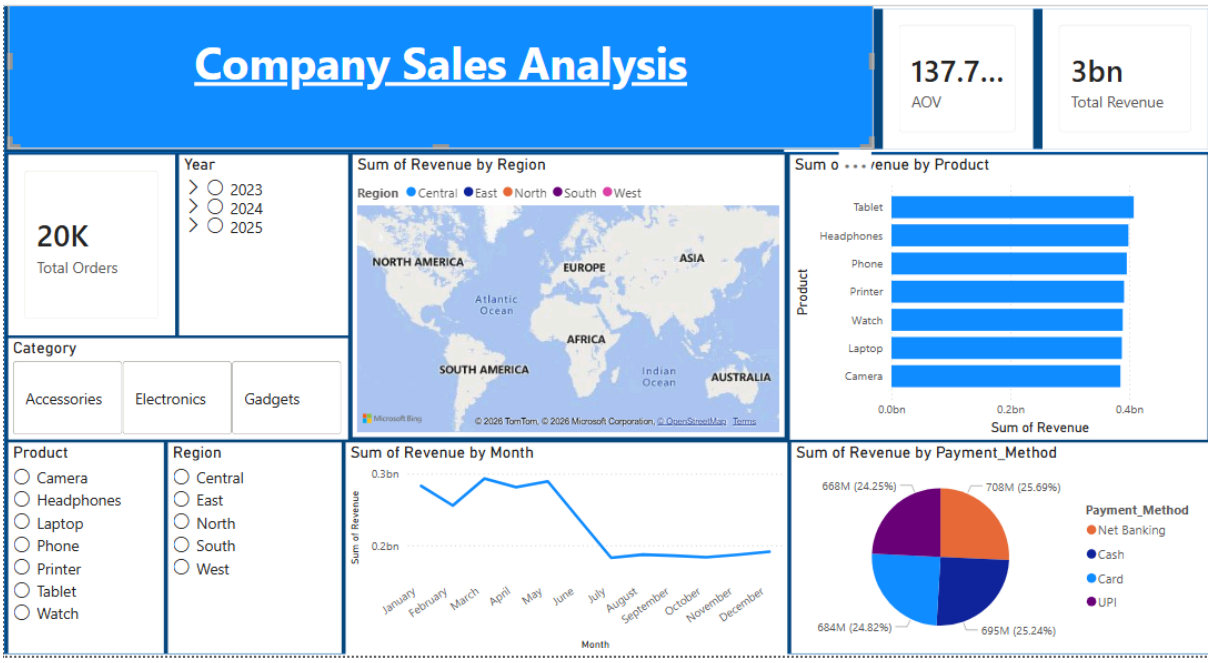
Payment_Method	Total_Transactions
Net Banking	5077
Cash	5044
Card	5023
UPI	4856

Monthly Sales Trend: Used to analyze monthly business performance and trends.

Month	Monthly_Revenue
2023-01	96159798
2023-02	88710224
2023-03	95325313
2023-04	95518291
2023-05	95095573
2023-06	87147449
2023-07	99596535
2023-08	89479475
2023-09	94625457
2023-10	90534037
2023-11	90162842

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Key Insights

● Net Banking is most used payment method
● Tablet generates highest revenue
● West region shows highest sales
● All categories contributed almost equally to total revenue.
● Digital payments (Net Banking, Card) were most preferred.
● Sales showed monthly fluctuations mostly in 2023 and 2024 and end of june 2025 it goes to its lowest point

7.Business Recommendations

● Focus marketing on top-selling products (Laptop, Phone)
● Improve sales in low-performing regions.
● Promote UPI and digital payments through offers.
● Maintain stock based on monthly demand trend